



INTRODUCTORY CONCEPTS OF PROBABILITY

PROBLEM SHEETS

Dr. Adebayo F. A.

Adebayof@ub.ac.bw
Block 240 – Office 226

Department of Statistics
University of Botswana

January, 2025

Problem Sheet 1

1. Let $U = \{1, 2, 3, \dots, 10\}$. Three subsets of U ; A , B and C are given by:

$$A = \{1, 2, 3, \dots, 8\} \quad B = \{2, 3, 4\} \quad C = \{1, 3, 5, 7, 9\}$$

- (a) Obtain the set $\bar{A}$, $B \cup \bar{C}$ and $A \cap (\bar{B} \cup C)$.
(b) Is it true that $\bar{B} \subset A \cap C$?
2. Draw a series of Venn diagrams representing three set, and shade in the following areas.

- (a) $\bar{A} \cap B \cap C$
(b) $(B \cap A) \cup (A \cap C)$
(c) $\overline{A \cup B \cup C}$
(d) $(A \cup B \cup C) \cap (\bar{B} \cap C)$

3. Let S denote the set of all companies listed on the Botswana Stock Exchange. Define the following sets as:

$$\begin{aligned} A &= \{x \mid x \text{ is in the mining sector}\}, \\ B &= \{x \mid x \text{ has an annual turnover exceeding P10 million}\}, \\ C &= \{x \mid x \text{ has a financial year ending in June}\}, \\ D &= \{x \mid x \text{ share price is higher now than six months ago}\}. \end{aligned}$$

Describe in words the following sets:

- (a) $A \cup B$
(b) $\overline{B \cap C}$
(c) $B \cap (C \cup A)$
(d) $(\overline{B \cap A}) \cup (C \cap D)$

4. If A is a set, let $\bar{A}$ denote the complement of A . Show the following

- (a) $(A \cap \bar{B}) \subset \overline{A \cap B}$
(b) $A - B = A$ if and only if $B - A = B$

5. Let A and B be two sets defined in the universal set, U . Depict the following sets in Venn diagrams:

- (a) $C = (A \cap \bar{B}) \cup (\bar{A} \cap B)$
(b) $D = (\overline{A \cap B}) \cup (A \cap B)$
(c) What can you say about the set C and D ?

Problem Sheet 2

1. A breakfast cereal manufacturer packs one of five pictures (A, B, C, D, E) in each box of cereal. If you buy two boxes, what is the sample space for the random experiment whose outcome is the two pictures in the boxes?
2. A small town has three grocery stores (Choppies, Bakgatla and SaveRite). Four ladies living in this town each randomly and independently pick a store in which to shop. Give the sample space of the experiment which consists of the selection of the stores by the ladies. Then define the events:

 A: all the ladies choose Choppies
 B: half of the ladies choose Choppies and half choose Bakgatla
 C: all the stores are chosen by at least one lady.
3. Let A and B be two events in the sample space S . Write down an expression for each of the following events, express their probabilities in terms of $\mathbb{P}(A)$, $\mathbb{P}(B)$ and $\mathbb{P}(A \cap B)$, and evaluate their probabilities if $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.4$ and $\mathbb{P}(A \cap B) = 0.2$:
 - (a) either A or B occurs
 - (b) A does not occur
 - (c) neither A nor B occurs
 - (d) exactly one A or B occurs
4. Show that $\mathbb{P}(A \cup \bar{B}) = 1 - [\mathbb{P}(B) - \mathbb{P}(A \cap B)]$.
5. If A , B and C are events in a sample space S , which of the following statements are impossible?
 - (a) $\mathbb{P}(A) = 0.7$, $\mathbb{P}(B) = 0.9$, $\mathbb{P}(C) = 0.3$ and $\mathbb{P}(A \cap B) = 0.4$.
 - (b) $\mathbb{P}(A) = 0.2$, $\mathbb{P}(B) = 0.5$, $\mathbb{P}(C) = 0.3$ and $\mathbb{P}(A \cap B) = 0.25$.
 - (c) $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.7$, $\mathbb{P}(C) = 0.8$ and $\mathbb{P}(A \cap B) = 0.1$.
 - (d) $\mathbb{P}(A) = 0.8$, $\mathbb{P}(B) = 0.4$, $\mathbb{P}(C) = 0.5$ and $\mathbb{P}(A \cup B) = 0.7$.

Problem Sheet 3

1. What is the probability that an 8-digit cellphone number has no repeated digits. The number can only start with the number 7 and 0 is not allowed as the second digit.
2. A pack of playing cards is being used by four players for a game of bridge, so each is dealt 13 cards. The **King**, **Queen** and **Jack** are referred to as *picture cards*. Find the probability that a bridge hand (13 cards) contains
 - (a) 3 spades, 4 diamonds, 1 heart and 5 clubs;
 - (b) 3 aces and 4 picture cards.
3. Suppose there are n people in a room.
 - (a) What is the probability that at least two are born in the same month? Assume all the months are equal.
 - (b) What is the probability if $n = 13$?
4. A class of 75 students is to be divided into three tutorial groups of sizes 25, 30 and 20 respectively.
 - (a) In how many ways can this be done?
 - (b) Suppose there are two brothers in this class. What is the probability that they are in the same tutorial class?
5. In how many ways can a student answer a 10-question T/F test if
 - (a) he marks half the questions **TRUE** and the other half **FALSE**?
 - (b) he marks no two consecutive answers the same?

Problem Sheet 4

1. The probability that a student passes Probability is 0.8 if she studies for the exam and 0.3 if she does not study. If 60% of the class studied for the exam, and a student picked a random from the class has passed, what is the probability that she studied for the exam?
2. Two machines are producing the same item. The previous week, **Machine A** produced 40% of the total output, and **Machine B** the remainder. On average, 10% of the items produced by **Machine A** were defective, and 4% of the items produced by **Machine B** were defective.
 - (a) What proportion of the previous week's entire production was defective?
 - (b) If an item selected at random from the combined output is found to be defective, what is the probability it came from **Machine A**?
3. An X-ray test is used to detect a disease that occurs, initially without any symptoms, in 3% of the population. The test has the following error rates:
 - 7% of people who are disease free have a positive reaction;
 - 2% of the people who have the disease have a negative reaction.A large number of people are screened at random using the test, and those with positive reaction are taken for further examination.
 - (a) What proportion of people who have disease are correctly diagnosed?
 - (b) What proportion of people with a positive reaction actually have the disease?
 - (c) What proportion of people with negative reaction actually have the disease?
 - (d) What proportion of the test conducted give incorrect diagnosis?
4. If A and B are events such that $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B|A) = 0.3$ and $\mathbb{P}(A \cup B) = 0.72$. Are A and B independent, mutually exclusive or both?
5. Show that if events A and B are independent, then the following pairs of events are also independent.
 - (a) A and $\bar{B}$
 - (b) $\bar{A}$ and $\bar{B}$.

Problem Sheet 5

1. State whether the following random variables are discrete or continuous.
 - (a) The time required to answer this question.
 - (b) The number of words in any random from the library.
 - (c) The maximum temperature recorded at Sir Seretse Khama International Airport.
 - (d) The number of application letters before successfully landing a job.
 - (e) The proportion of students passing STA122.
2. Suppose that a die is rolled twice. What are the possible values and their corresponding probabilities that the following random variables can take on:
 - (a) the maximum value to appear in both rolls;
 - (b) the sum of the values obtained from the two rolls;
 - (c) the value of the first roll minus the value of the second roll?
3. Two balls are chosen at random from an urn containing 6 white, 2 black and 2 red balls. Suppose that you win P20.00 for each black ball selected and you lose P10.00 for each white ball selected. Let X be a random variable denoting your winnings. What are the possible values of X , and what are the probabilities associated with each value?
4. Consider a multiple-choice test with three (3) possible answers for each of the five (5) questions. Let X be a random variable representing whether a student pass or fail the test given that the student just guessed the answers. What are possible values of X , and the probabilities associated with each value?
5. A small pool building company is **equally likely** to complete 2 or 3 contracts each month. The company receives between 1 and 4 contracts to build pools each month, with probabilities $\mathbb{P}(1) = 0.1$, $\mathbb{P}(2) = 0.2$, $\mathbb{P}(3) = 0.5$ and $\mathbb{P}(4) = 0.2$. At the beginning of this month the company has two contracts carried forward from the previous month. Let X be a random variable representing the number of contracts to be carried forward to the next month. What are possible values of X , and the probabilities associated with each value? Assume that the number of contracts is independent of the number of pools completed, and that the contracts are made at the beginning of the month.

Problem Sheet 6

1. Which of the following functions satisfy the conditions for being a probability mass function?

(a)

$$p(x) = \begin{cases} 0.25(0.75)^x & x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Hint: use a geometric series, $\sum_{i=0}^{\infty} a^i = 1/(1-a)$.

(b)

$$p(x) = \begin{cases} \frac{1}{5} \binom{2x-3}{x} & x = 0, 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

(c)

$$p(x) = \begin{cases} \frac{\binom{5}{x} \binom{4}{x-2}}{36} & x = 0, 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

2. Consider the following function,

$$p(x) = \begin{cases} \frac{k}{x!} & x = 0, 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) For what value of k would $p(x)$ be a probability mass function.
(b) Find $\mathbb{P}(X < 2)$.

3. A random variable, X has a probability mass function given by

$$p(x) = \begin{cases} \frac{12}{25^x} & x = 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

Plot the bar graph corresponding to this probability mass function.

4. If the cumulative distribution function of X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} & 0 \leq x < 1 \\ \frac{3}{5} & 1 \leq x < 2 \\ \frac{4}{5} & 2 \leq x < 3 \\ \frac{9}{10} & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

calculate the probability mass function of X .

Problem Sheet 7

1. For the probability density function

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

find the number α such that the probability that $X < \alpha$ is three times the probability that $X \geq \alpha$.

2. A random variable X has probability density function

$$f(x) = \begin{cases} \exp(-x) & 0 \leq x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the number t such that $\mathbb{P}(X < t) = 0.5$. Note that derivative of $\exp(-x)$ w.r.t x is $-\exp(-x)$.

3. Suppose that the random variable X is equal to the number of hits obtained by a certain softball player in his next 3 bats. If $\mathbb{P}(X = 1) = 0.3$, $\mathbb{P}(X = 2) = 0.2$ and $\mathbb{P}(X = 0) = 3\mathbb{P}(X = 3)$, find $\mathbb{E}(X)$ and $\mathbb{V}(X)$.
4. A small pool building company is equally likely to complete 2 or 3 contracts each month. The company receives between 1 and 4 contracts to build pools each month, with probabilities:

$$\mathbb{P}(1) = 0.1; \mathbb{P}(2) = 0.2; \mathbb{P}(3) = 0.5 \text{ and } \mathbb{P}(4) = 0.2.$$

At the beginning of this month the company has two contracts carried forward from the previous month. Let X be a random variable representing the number of contracts to be carried forward to the next month. Find the expected value and variance of X .

5. The density function for a random variable X is given by

$$f(x) = \begin{cases} \frac{3}{4}(kx - x^2) & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Determine the value of k and find the expected value and variance of X .

Problem Sheet 8

1. A **true-false** test is given for five questions. To pass, you need at least four (4) questions to be correct. If you guess each answer, what is the probability that you'd pass the test?
2. Airplane engines operate independently in flight and fail with probability of 0.1. A plane makes a successful flight if at most half of its engines fail. Determine the probability of a successful flight for two-engined and four-engined planes.
3. A road is constructed so that the right-hand turn lane at an intersection has a capacity of three (3) cars. Suppose that 30% of cars approaching the intersection want to turn right. If a string of 15 cars approaches the intersection, what is the probability that the lane will be insufficiently large to hold all the cars waiting to turn right?
4. An ice-cream vendor's sales follow a Poisson distribution with an average rate of 10 sales per hour.
 - (a) What is the probability that he sells at least one ice-cream in his first hour of operation?
 - (b) How much should his stock of ice-cream be at any point in time if he wants to be at least 95% sure that he does not run out of ice-cream in the following hour?
5. A pump fails, on the average, once in every 500 hours of operation. What is the probability that it would fail exactly three (3) times in 2000 hours of operation?

Problem Sheet 9

1. A TV manufacturer is supplied with a certain component by a specialist producer. Each incoming consignment of components is subjected to the following quality control procedure. A random sample of 10 components is individually tested. If there is one or more defective components among the 10 tested, the entire consignment is rejected.
 - (a) What are the probabilities of a consignment being rejected if the true proportions of defective components are
 - (i) 1%
 - (ii) 10%
 - (iii) 30%
 - (b) If a sample of 20 components (instead of 10) were tested, and the consignment rejected if two or more proved defective, calculate the probabilities of rejecting a consignment for the same proportions of defective components.
 - (c) Which quality control procedure do you think is better?
2. The average demand on a factory store for a certain electric motor is 8 per week. When the storeman places an order for these motors, delivery takes one week. If the demand for motors has a Poisson distribution, how low can the storeman allow his stock to fall before ordering a new supply if he wants to be at least 95% sure of meeting all requirements while waiting for his new supply to arrive?
3. What is the probability that a fair die is tossed 10 times before 6 appears
 - (a) for the first time;
 - (b) for the 6th time?
4. A presidential candidate needs to collect 50 signatures before she can be nominated, **and no, I'm not talking about Toitoti!** If the probability that a delegate approached at random will give a signature is 0.15, what is the probability that 300 delegates need to be approached before the 50th signature is collected?
5. A tuckshop has 10 cartons of milk left, of which three are sour. Unaware of this, you ask for four cartons of milk. If the four cartons are selected at random, what is the probability that at least half of these are sour?

Problem Sheet 10

1. A radio station announces the time every 15 minutes between midnight and 06h00. If you wake up at random in the early hours of the morning and switch on the radio, what is the probability that you have to wait less than 5 minutes to find out the time?
2. The life of an electronic device is known to have an exponential distribution with parameter $\lambda = 1/1000$.
 - (a) What is the probability that the device lasts less than 1000 hours?
 - (b) What is the probability it will last more than 1200 hours?
 - (c) If three such devices are taken at random, what is the probability that one will last less than 800 hours?
3. Customers arrive at a restaurant at the rate of 30 per hour during lunch time. If a minute has already passed by since the last customer arrived, what is the probability that it is at least another minute before the next customer arrives?
HINT: use conditional probability.
4. If X is distributed as a normal variable with mean $\mu = 3$ and variance $\sigma^2 = 4$, evaluate the following probabilities
 - (a) $\mathbb{P}(X < 4)$
 - (b) $\mathbb{P}(|X| < 6)$
 - (c) Find a such that $\mathbb{P}(X < a) = 0.80$.
5. If X has a normal distribution, and if $\mathbb{P}(X < 10) = 0.8413$, what is the value of the mean if the distribution is known to have variance $\sigma^2 = 16$?
6. The maximum (stated) load of passenger lift is 10 passengers or 600kg. If the masses of people using the lift can be considered to be normally distributed with mean 65kg and standard deviation 15kg, how often will the combined mass of 10 passengers exceed the 600kg limit?